

Alpilla 600 MW Combined Cycle Power Plant

Civil and Structural Design Criteria

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A	IFR	2026-12-14	First issue for Owner approval (Basic Design Package)	EPCE	EPC civil lead	EPC engineering manager

1. Purpose and scope

1.1 This document defines the civil and structural design criteria for the Alpilla 600 MW combined cycle power plant (1 x 1 multi-shaft, power island by IEC, balance of plant by IEPC). It applies to the design of all foundations, buildings, structures, pipe racks and supports, roads, drainage and the marine works (seawater intake and outfall) and is the basis for ALP-EPC-00000-CV-CAL-0001, CV-SPC-0001..0005 and the civil / structural drawings.

1.2 Where TBDY 2018 and the Eurocodes give different requirements for the same check, the more onerous shall govern; for seismic actions and detailing TBDY 2018 governs, for all other actions and for steel and concrete member design the Eurocodes (EN 1990 to EN 1993 with the Turkish National Annexes) govern, TS 500 being used only where TBDY 2018 refers to it or where it is more onerous for the same check.

1.3 This document is a design basis, not a specification. Values marked preliminary shall be confirmed by the site-specific seismic hazard study (ALP-EPC-00000-CV-STU-0001), the geotechnical interpretation report (ALP-EPC-00000-CV-RPT-0001) and the OEM foundation and interface data before the first issue for construction.

2. Governing requirements

ER-13.01: Turbine and generator foundations designed for dynamic loads with separated foundations from buildings; settlement limits per OEM.

ER-13.02: Turbine hall, control and electrical building, administration building, workshop/warehouse, water treatment building, gatehouse, fire station.

ER-13.03: Structural steel per EN 1993 with seismic detailing, hot-dip galvanised outdoors (EN ISO 1461).

ER-13.04: Concrete per EN 1992 / TS 500 with exposure class XS for marine-exposed elements; minimum 50-year durability.

ER-13.05: Internal asphalt roads for 40 t axle loads to the heaviest delivery route, surface water drainage for ER-02.05 rainfall.

ER-13.06: Intake and outfall pipelines (HDPE or GRP) with ballast and scour protection, designed for 50-year wave conditions.

ER-02.05: Basic wind speed 30 m/s (10 min mean, 10 m height, 50-year return); ground snow load 0.75 kN/m²; design rainfall intensity 110 mm/h (10 min, 25-year return) for drainage design.

ER-02.06: The Site is in a high seismic hazard region. Structures and equipment shall be designed per the Turkish Building Earthquake Code (TBDY 2018) with a site-specific hazard study; design earthquake DD-2 (475-year) peak ground acceleration 0.45 g; building importance factor 1.5 for power block structures; functional recovery of critical systems after DD-3 events.

ER-02.07: Owner-provided baseline investigation indicates alluvial deposits over weathered rock at 18 m to 25 m depth, groundwater at +13.5 m and liquefaction potential in the upper sand layers. The Contractor shall perform confirmatory investigations and select foundation solutions (piling or ground improvement) accordingly.

ER-02.08: Outdoor equipment and structures shall be designed for corrosivity category C5 (ISO 12944-2) with a coating system of high durability (H, more than 15 years to first major maintenance).

ER-02.09: Finished floor levels of all buildings and equipment foundations shall be at least 0.5 m above the 100-year coastal flood level including a sea-level-rise allowance of 0.5 m.

3. Codes and standards

3.1 The following codes govern, in this order for a conflict between them: (1) Turkish legislation (Zoning Law, Building Earthquake Code TBDY 2018, Fire Regulation); (2) the Employer's Requirements; (3) TBDY 2018 for seismic actions and seismic detailing; (4) EN 1990 with EN 1991 (actions), EN 1992 (concrete), EN 1993 (steel) with National Annexes; (5) TS 500 where cited above; (6) ACI 351.3R for dynamic equipment foundations; (7) ISO 20816-2 for vibration evaluation; (8) EN ISO 1461 and ISO 12944-2 for corrosion protection.

3.2 Codes not in the project reference register but named in this document (EN 1997 geotechnical design, EN 206 concrete, EN 1994, EN 1998 as cross-check, EN 12699 / EN 1536 piling execution, TS 708, TS EN 197-1, EN 10025, EN 1090, ISO 4354 or EN 1991-1-4 wind, API 650 Annex B) shall be registered before they are cited in a specification.

4. Site and geotechnical data

Parameter	Value	Condition	Status	Basis
100-year coastal still water level	0.9 m	national datum; +1.40 m with 0.5 m sea-level rise; floor levels $\geq$ +1.90 m (ER-02.09); shoreline run-up +4.8 m	confirmed	SRC-OWNR-0002, ER-02.09
Atmospheric corrosivity category	C5 (ISO 12944-2), durability H -		confirmed	ER-02.08
Basic wind speed	30 m/s	10 min mean, 10 m, 50-year return	confirmed	ER-02.05
Design rainfall intensity	110 mm/h	10 min, 25-year return	confirmed	ER-02.05
Distance to heavy-lift port	18 km by sea	400 t quay crane	confirmed	SRC-OWNR-0005
Ground snow load	0.75 kN/m ²		confirmed	ER-02.05
Groundwater level	13.5 m a.s.l.	brackish, chloride 3,000-8,000 mg/l	confirmed	SRC-OWNR-0008
Heavy transport road limit	120 t gross	two bridges on the state road	confirmed	SRC-OWNR-0005, ER-02.14
Height datum	TUDKA-99 orthometric heights (m a.s.l.)		confirmed	SRC-EPCE-0011
Plant grid origin in UTM zone 35N (E 0 / N 0)	E 541,250.000 m / N 4,532,480.000 m	TUREF (ITRF96), south-west plot corner	confirmed	SRC-EPCE-0011
Platform level acceptance tolerance	0.15 m	Owner-levelled platform +15.00 m	confirmed	DEC-OWNR-0002
Plot dimensions	400 x 300 m	12.0 ha; usable about 10.6 ha	confirmed	SRC-OWNR-0005, ER-02.01
Rotation plant grid to UTM grid	-0.3281 deg	plant north = true north; grid convergence at site	confirmed	SRC-EPCE-0011
Scale factor plant grid to UTM	0.999612 -	combined UTM point scale and height factor	confirmed	SRC-EPCE-0011
Seismic design PGA (DD-2, 475-year)	0.45 g	TBDY 2018, site-specific hazard study	confirmed	ER-02.06
Seismic site class	ZD (Vs30 = 240 m/s) -	TBDY 2018	confirmed	SRC-OWNR-0008
Site grade level	15 m a.s.l.		confirmed	ER-02.01

4.1 The Site is at grade +15.00 m a.s.l. on Owner-placed engineered fill (1.5 m to 3.0 m, compacted to 95 % Standard Proctor). The typical profile per ALP-OWN-GEO-001 is: 0 m to 2.5 m engineered fill; 2.5 m to 9 m loose to medium dense silty sand (SPT N 6 to 18), liquefiable under the DD-2 earthquake; 9 m to 18 m soft to firm compressible clay (cu 30 to 70 kPa); 18 m to 25 m dense sand and gravel (SPT N > 40) where present; weathered sandstone / marl (UCS 5 MPa to 25 MPa) below, with the top of rock between about 18 m and 25 m. Depths are below grade.

4.2 Groundwater is at +13.5 m a.s.l. (1.5 m below grade), brackish with chloride 3,000 mg/l to 8,000 mg/l, and shall be treated as aggressive to concrete (exposure XS2 and XA2 for buried elements). The design groundwater level for buoyancy and uplift checks of basements, pits and tanks shall be grade (+15.00 m); for settlement and effective stress checks the measured level +13.5 m applies with a sensitivity check at +14.5 m.

4.3 The Owner geotechnical baseline (12 boreholes to 30-45 m, 20 CPTu, 6 MASW) is information only. The Contractor shall carry out a confirmatory investigation before foundation design: at least one borehole with SPT and one CPTu to refusal within the footprint of each of the GT, ST, HRSG, stack, GSU transformer, main building and tank foundations, boreholes to at least 5 m into weathered rock at the pile positions, cross-hole or down-hole shear wave velocity at the turbine foundations, and consolidation and cyclic tests on the clay and sand layers. The results shall be issued in the geotechnical interpretation report ALP-EPC-00000-CV-RPT-0001.

5. Levels, datum and coordinates

5.1 All levels are orthometric heights in the TUDKA-99 datum and all coordinates are in the plant grid defined by DP-EPCE-0001 to DP-EPCE-0004 (site block, section 4). The plant grid rotation and scale factors shall be applied in the setting-out report ALP-EPC-00000-CV-RPT-0003.

5.2 The nominal platform level is +15.00 m. The Owner platform is accepted within +/- 0.15 m (DP-EPCE-0008). Finished floor levels and road levels shall be set on the as-surveyed platform with local trimming only; structural fill needed below +14.85 m is not Contractor scope of cost (paid by the Owner at the Price Schedule unit rate against a joint survey) as agreed in TQ-039 and DEC-OWNR-0002. The south edge strip (N 0 to 25 m) and the old warehouse foundations shall be resurveyed jointly before setting out.

Item	Value	Unit	Status	Basis / remarks
Finished floor level of buildings (ground floor)	+15.30 m (0.30 m above platform)		assumption	ER-02.09, DP-CASE-0001; Terrace above roads; ER-02.09 minimum +1.90 m met (TQ-041)
Top of equipment foundations and pedestals outdoors	not less than +15.20 m (0.20 m above platform)		assumption	ER-02.09
Road and paved area level	+15.00 m at the crown, minimum fall 0.5 %		assumption	DP-CASE-0001
Minimum floor level above 100-year coastal flood level	floors at least 0.5 m above the 100-year coastal still water level plus 0.5 m sea-level rise, i.e. not below +1.90 m		confirmed	ER-02.09, DP-EPCE-0015, TQ-041; 100-year still water level per the design parameter (TQ-041)

5.3 The plant platform (+15.00 m) is far above the ER-02.09 minimum floor level of +1.90 m (TQ-041). Structures at the shoreline (intake head works, CW pump house, outfall seal pit) shall have their floors at least at +5.3 m (50-year wave run-up +4.8 m plus 0.5 m, TQ-041); parts below these levels (pits, sumps, intake and outfall works) shall be watertight and designed for flooding. The check is documented in CV-CAL-0001.

6. Design loads

6.1 Design life of the plant is per DP-CASE-0034; the structural design working life is 50 years (EN 1990 category 5) for buildings and main structures and 30 years minimum for replaceable items (pipe supports, cladding, non-structural elements), the concrete durability design being for 50 years (ER-13.04). Consequence class CC3 (EN 1990) applies to the power block structures, turbine foundations and the control and electrical building; CC2 to other structures.

6.2 Dead loads: self weight from actual densities (reinforced concrete 25 kN/m³, structural steel 78.5 kN/m³, engineered fill 20 kN/m³); permanent equipment loads, piping, cable trays, insulation, cladding and fireproofing shall be taken from the equipment weights in the database and the vendor drawings; where not yet known an allowance of 1.5 kN/m² on platforms and 0.5 kN/m² on roofs shall be added and tracked as an open item.

Item	Value	Unit	Status	Basis / remarks
Imposed load - turbine hall operating floor and laydown	20 kN/m ² (or maintenance load case from OEM if greater)		assumption	REF-CONS-0087; Laydown areas per OEM outage plan
Imposed load - control and electronic rooms (raised floor)	5.0 kN/m ²		assumption	REF-CONS-0087
Imposed load - MV / LV switchgear and battery rooms	10 kN/m ² or actual equipment weight if greater		assumption	REF-CONS-0087; Cable basement and floor openings checked separately
Imposed load - offices, meeting rooms, gatehouse	3.0 kN/m ²		assumption	REF-CONS-0087
Imposed load - workshop and warehouse floors	10 kN/m ² (plus forklift 3 t axle)		assumption	REF-CONS-0087
Imposed load - stairs, corridors and escape routes	4.0 kN/m ²		assumption	REF-CONS-0087
Imposed load - grating platforms and walkways	3.0 kN/m ² and 5 kN/m ² at maintenance laydown zones; 1.5 kN point load		assumption	REF-CONS-0087
Imposed load - roofs not accessible	0.4 kN/m ² (category H) plus 1.0 kN concentrated		assumption	REF-CONS-0087

6.3 Wind loads shall be per EN 1991-1-4 for the basic wind speed of DP-CASE-0004 (50-year return, 10 min mean at 10 m), terrain category 0 (open sea fetch) for the sea-facing sector and category II otherwise, with the topography factor taken as 1.0 (flat platform). Dynamic response and vortex shedding shall be checked for the stack (65 m, DP-CASE-0043), the HRSG casing and other slender structures; the HRSG and the stack are vendor-supplied and the wind data and combinations of this document apply as interface.

6.4 Snow load: characteristic ground snow load per DP-CASE-0005, with the shape and exposure coefficients of EN 1991-1-3; drift accumulations shall be included at roof steps and at the turbine hall / auxiliary bay junctions. Rain: roof drainage shall be sized for the rainfall intensity of DP-CASE-0006 and the roof shall carry the ponding load of blocked primary drains up to the level of the overflow (secondary drainage of the same capacity, not counted in the primary capacity).

6.5 Temperature actions: uniform temperature range of exposed steel structures from -15 degC to +45 degC (survival range DP-CASE-0009), solar heating of +20 K for dark-coloured surfaces in addition to the ambient; thermal expansion of pipe racks shall follow the piping stress analysis with anchor and guide loads supplied by the piping discipline.

6.6 Crane and lifting loads shall be per EN 1991-3 and EN 13001 with dynamic factors of the crane class; the turbine hall overhead cranes shall be designed for the heaviest maintenance lift given by the OEM (rotor, casing or generator stator) with the vertical load on the runway beams, lateral loads and buffer forces combined with wind and seismic loads.

6.7 Pipe, vessel and equipment loads on supports shall be taken as empty, operating and hydrotest weight, with thrust, friction, anchor and safety valve reaction loads supplied by piping stress; the hydrotest load shall be combined with 30 % of the wind load and no seismic load, and the operating load with full wind and full seismic loads.

7. Seismic design

7.1 Seismic design shall be per TBDY 2018 for the design earthquake DD-2 (475-year), site peak ground acceleration 0.45 g and site class ZD ($V_{s30} = 240$ m/s). The site-specific hazard study CV-STU-0001 shall define the design spectrum; if the site-specific spectrum is lower than the code spectrum for the ZD class, the code spectrum shall not be reduced by more than the limit TBDY 2018 permits (80 % of the code value at any period).

7.2 Building importance class: BIS 1 (importance factor $I = 1.5$) shall be used for power block structures, the control and electrical building, the GIS and switchyard structures, fire water and fire station facilities, and the foundations and supports of critical equipment. Other buildings (administration, workshop, warehouse, gatehouse) shall be designed as BIS 3 ($I = 1.0$) unless they house emergency functions.

7.3 Behaviour factors: the ductility level shall be high for all lateral load resisting systems of power block structures; the response modification factor R and the overstrength factor D shall be taken from TBDY 2018 for the chosen system, with R limited to 6 for steel structures of the power block and to 5 for reinforced concrete frames without shear walls (project limit to keep drift under control), and $R = 1.5$ for non-building structures that are inverted-pendulum or that have no redundancy (stack, tanks on ring walls per the tank code).

7.4 Functional recovery: critical systems (control building, electrical building, emergency diesel generator, fire water system, battery rooms, gas station) shall remain functional after a DD-3 event; for this condition the structures shall remain elastic, the interstorey drift shall not exceed 0.5 % of the storey height and the equipment supports and anchorages shall remain elastic.

7.5 Non-structural elements and equipment: seismic forces on equipment, piping supports, cable trays, cladding and suspended items shall be calculated per TBDY 2018 chapter 6 with the floor acceleration amplification, R_p and a_p factors given there and the importance factor $I = 1.5$ for critical equipment. Equipment anchorage shall be verified for the DD-2 forces with overstrength; vendors shall supply anchor bolt loads for the seismic case and confirm seismic qualification of their equipment (analysis or test) for 0.45 g at the foundation level.

7.6 Seismic separation: buildings and structures with different dynamic behaviour shall be separated by seismic joints with a width not less than the sum of the DD-2 displacements (with a minimum of 100 mm for pipe rack to building and 150 mm for building to building); pipe crossings across joints shall be designed for the relative displacement. Turbine and generator foundations shall be independent of building foundations and floors (ER-13.01).

7.7 Liquefaction: the potential shall be assessed per TBDY 2018 chapter 16 for the DD-2 event using the CPTu and SPT data; the loose silty sand layer (2.5 m to 9 m below grade) is expected to liquefy. Foundations shall either transfer the loads below the liquefiable layer (piles) or the layer shall be improved (stone columns, deep soil mixing) so that the liquefaction safety factor is not less than 1.2 under DD-2; post-liquefaction settlement and lateral spreading shall be included in the pile design (kinematic loads, loss of skin friction in the liquefied layer).

8. Load combinations

8.1 Ultimate limit state combinations shall be per EN 1990 equation 6.10 (or 6.10a / 6.10b), with the partial factors and ψ_i factors of the National Annex and category E / F imposed load factors for industrial areas. The seismic combination shall be per TBDY 2018 with EN 1990 equation 6.12b: $G + \psi_{i2} \times Q + E$, $\psi_{i2} = 0.6$ for storage, laydown and equipment areas.

8.2 The following project load cases shall be combined: dead, imposed, equipment operating, equipment empty and hydrotest, wind (four directions), seismic (two horizontal components with 100 % / 30 % combination and the vertical component where TBDY requires), snow, thermal, crane, pipe reactions, fault loads of the machines (short-circuit torque, blade loss, bearing failure per OEM) and construction / erection loads. Fault load cases of the OEM shall not be combined with seismic loads.

8.3 Serviceability combinations (characteristic, frequent, quasi-permanent) shall be per EN 1990; the crack width and deflection limits of sections 9, 10 and 12 apply to the quasi-permanent combination unless stated otherwise.

9. Dynamic machine foundations

9.1 Foundations for the gas turbine, steam turbine, generators and other rotating equipment above 500 kW (boiler feed pumps, CW pumps, condensate pumps, compressors) shall be designed per ACI 351.3R and the OEM foundation requirements. The gas turbine, generator and steam turbine foundations shall be table-top or block foundations on piles or improved ground, separated from building foundations and floors by isolation joints (ER-13.01).

9.2 Frequency separation: the natural frequencies of the coupled foundation - soil system in the vertical, horizontal and rocking modes and of the table-top structure shall differ by at least 20 % from the operating frequency (50 Hz, 3000 rpm), from twice the operating frequency where the OEM gives excitation at that frequency, and from every other excitation frequency of the OEM; the resonance zone shall not be crossed at a speed where sustained vibration is possible (transient crossing during run-up and coast-down shall be checked for amplitude).

9.3 Soil dynamic parameters shall be derived from the shear wave velocity measurements at the foundation. The dynamic analysis shall be run for the best-estimate shear modulus and for lower and upper bounds (0.5 and 2.0 times the best estimate, project assumption) and, for piled foundations, with the dynamic pile group stiffness; the frequency separation shall be met for all bounds.

9.4 Vibration limits: the vibration of the foundation under the OEM dynamic loads shall not exceed the OEM limits; where the OEM gives no limit the amplitude of the top of the foundation shall not exceed a peak velocity of 2.0 mm/s at the operating frequency (project limit). The machine vibration shall be evaluated per ISO 20816-2 (zone A / B for new machines; alarm at the zone B / C boundary and trip at C / D unless the OEM sets other levels) and the foundation shall not cause a change of the zone.

9.5 Foundation mass and geometry: for block foundations and for the mat under a table-top the mass shall not be less than 3 times the mass of the supported machine (rotating machine including generator) unless the dynamic analysis demonstrates the frequency and amplitude criteria; the centre of gravity of foundation and machine shall not deviate by more than 5 % of the foundation dimension in plan (eccentricity limit).

9.6 Concrete of dynamic foundations: minimum class C35/45 with the exposure requirements of section 10; the minimum reinforcement shall be 0.25 % of the section in each face and direction for massive blocks (thermal cracking control per EN 1992-3 where applicable); thermal control of mass concrete pours shall limit the core-to-surface temperature difference to 20 K and the maximum core temperature to 65 degC (delayed ettringite formation); embedded anchor bolts, sleeves and grout shall be per the OEM installation drawing.

9.7 Settlement of the machine foundations shall be limited to the OEM values; until the OEM values are known the differential settlement along the shaft train shall not exceed 1 : 2000 and the total settlement shall not exceed 15 mm. Levelling monuments shall be installed and read during construction, hydrotest of the HRSG, first fill of the condenser and during the first year of operation.

10. Concrete and reinforcement

10.1 Concrete shall be designed per EN 1992-1-1 with the Turkish National Annex and TBDY 2018 / TS 500 for seismic detailing, and shall be specified per EN 206 and TS EN 197-1. The exposure classes shall be: XS1 for above-ground elements exposed to airborne salt (all outdoor elements, given corrosivity C5 and chloride deposition of the site); XS3 for elements in the tidal and splash zone and within 2 m above the maximum sea level (intake and outfall structures, CW pumphouse pit walls); XS2 and XA2 for elements below the groundwater level (foundations, basements, pits, tank foundations); XC4 / XF1 are not decisive.

Item	Value	Unit	Status	Basis / remarks
Minimum concrete class - XS1 / XS2 elements	C35/45, w/c ≤ 0.45		assumption	ER-13.04, REF-CONS-0088
Minimum concrete class - XS3 elements	C40/50, w/c ≤ 0.40		assumption	ER-13.04, REF-CONS-0088
Cement type	CEM III/A 42.5 N (blast furnace slag ≥ 50 %) or CEM II/B-V with ≥ 6 % silica fume; no CEM I alone		assumption	ER-13.04, SRC-OWNR-0008; Sulphate and chloride resistance; heat of hydration
Chloride content of fresh concrete	≤ 0.10 % of cement mass (class CI 0.10 for reinforced concrete)		preliminary	REF-CONS-0088
Minimum nominal cover - XS1 elements	50 mm (structural class S4, 50 years, plus 10 mm tolerance)		assumption	ER-13.04, REF-CONS-0088
Minimum nominal cover - XS3 elements	65 mm		assumption	ER-13.04, REF-CONS-0088

Item	Value	Unit	Status	Basis / remarks
Minimum nominal cover - cast against blinding	65 mm (75 mm for cast-in-place piles and for concrete cast against soil)		assumption	REF-CONS-0088
Crack width limit (quasi-permanent combination)	0.30 mm for XS1 above ground; 0.20 mm for XS3, XS2 / XA2 and liquid retaining elements		preliminary	REF-CONS-0088, ER-13.04
Reinforcement	Ribbed B500C (ductility class C, TS 708), weldable; no reliance on coated or galvanised bars; corrosion-inhibiting admixture for XS3 elements		assumption	REF-CONS-0088, REF-CONS-0017, REF-CONS-0089

10.2 The 50-year durability of ER-13.04 shall be demonstrated by the cover, concrete quality and crack width limits above and by a chloride ingress check per fib Model Code / EN 1992 for the XS3 elements. Concrete shall be cured for at least 7 days; the permeability (water penetration depth per EN 12390-8) shall not exceed 30 mm and the chloride migration coefficient (NT Build 492) shall not exceed 8×10^{-12} m²/s at 28 days for XS3 concrete.

10.3 Water-retaining structures (CW pumphouse, screens chamber, pits, sumps, tank farm recovery sump) shall be designed for leak tightness class 1 to 2 of EN 1992-3 and protected internally where the medium or the groundwater is aggressive (epoxy lining for oil sumps and chemical containment).

11. Structural steel and corrosion protection

11.1 Structural steel shall be designed per EN 1993-1-1 and EN 1993-1-8 with seismic detailing per TBDY 2018; fabrication and execution class EXC3 per EN 1090-2 for the power block structures and EXC2 for other structures; steel grades S355J2 for main members and S275JR for secondary members and grating supports; thick plates ($t > 30$ mm) and seismic energy dissipating members shall be tested for through-thickness properties (Z25) and Charpy J2 at -20 degC; the minimum service temperature is -15 degC (DP-CASE-0009).

11.2 Bolting: structural bolts M16 minimum, grade 8.8 hot-dip galvanised with matching nuts and washers; slip-critical and seismic connections grade 10.9 preloaded, galvanised or with a sherardised / zinc flake coating; anchor bolts galvanised, in sleeves to the OEM drawing. Welded connections in seismic force resisting systems shall use full penetration welds in the plastic hinge regions.

11.3 Corrosion protection: all outdoor structural steel shall be hot-dip galvanised per EN ISO 1461 (mean coating thickness not less than 85 microns for sections over 6 mm, 70 microns for 3 mm to 6 mm) as required by ER-13.03; the corrosivity category is C5 with durability H (DP-CASE-0003, more than 15 years to first major maintenance). Where galvanising is impractical (large welded assemblies, site-welded joints, steel in the splash zone, surfaces of members after erection damage), painting per ISO 12944 C5 high durability (or CX for the splash zone) with a duplex system shall be used per the painting and coating specification ALP-EPC-00000-ME-SPC-0002 and GE-SPC-0004; touch-up of galvanising damage with zinc-rich paint is required.

11.4 Structural steel inside buildings in the conditioned rooms (control, electrical, administration buildings) may be painted to ISO 12944 C3 durability M; steel in the turbine halls and other ventilated buildings shall be protected as outdoor steel because the ventilation air carries salt aerosol (chloride deposition up to the value of DP-OWNR-0007).

11.5 Fire protection of structural steel shall follow the fire resistance requirements of the Fire Regulation and NFPA 850 as determined in the fire protection concept (ALP-EPC-00000-HS-HAZ-0009): a minimum resistance of R 60 for the structures supporting or adjacent to LDO, gas station, transformers and the turbine hall and R 120 for the control and electrical building structure and for firewalls; passive fire protection shall be an intumescent coating compatible with the C5 system or concrete encasement.

12. Deflection and drift limits

Element	Limit	Remarks
Vertical deflection - floor beams (imposed load)	span / 300; total span / 250	[REF-CONS-0090]
Vertical deflection - roof purlins and cladding rails	span / 200 (imposed and snow)	[REF-CONS-0090]
Vertical deflection - cantilevers	length / 150	[REF-CONS-0090]
Crane runway beams	vertical span / 600; horizontal span / 500 (or the crane vendor limit if smaller)	[REF-CONS-0090, REF-CONS-0072]
Pipe rack beams	span / 300; smaller if the pipe stress or drainage slope requires	[REF-CONS-0090]
Horizontal deflection of frames under wind (serviceability)	height / 300 (industrial frames), height / 500 for frames carrying cranes or brittle cladding	[REF-CONS-0090]
Interstorey drift under DD-3 (functional recovery)	0.5 % of storey height	[ER-02.06, REF-CONS-0017]
Drift under DD-2 (life safety)	per TBDY 2018 limits ($\lambda \times \delta$ / h_s limit for the ductility level)	[REF-CONS-0017]

13. Foundations

13.1 Foundation solution: for all heavy and vibrating structures (GT, ST, generators, HRSG, stack, GSU transformers, condenser, main buildings, boiler feed and CW pumps, LDO and fire water tank rings) deep foundations shall be used: bored cast-in-place piles socketed into the weathered sandstone / marl at 18 m to 25 m depth, designed to EN 1997-1 and TBDY 2018 chapter 16; ground improvement (stone columns or deep soil mixing) is an option for lightly loaded structures, pipe racks, roads and small buildings, subject to demonstration of the liquefaction and settlement criteria.

13.2 Pile design: the axial capacity shall be based on the end bearing and socket resistance in the rock only; skin friction in the liquefiable sand and in the settling clay shall be neglected or treated as negative skin friction (downdrag) for the design earthquake and consolidation settlement of the fill; pile lateral capacity shall be checked for the inertial load combined with either kinematic loads from the soil (lateral spreading) or with soil stiffness degradation, whichever governs; group effects and pile cap stiffness shall be considered.

13.3 Pile testing: a preliminary static compression test on at least two piles per pile type (loaded to 2.0 times the working load) and a static lateral test shall be carried out before the production piles are ordered (ALP-EPC-00000-CV-TST-0001, CV-TRP-0001); working piles shall receive integrity tests (100 %) and static tests on at least 1 % of the piles; the pile design basis is issued in ALP-EPC-00000-CV-RPT-0001.

13.4 Pile concrete and reinforcement shall be XS2 / XA2 with the cover of section 10 and shall be placed with a polymer or bentonite support fluid and a tremie; sacrificial steel or increased cover shall be considered for the top part of the pile in the groundwater; the pile head shall be embedded in the cap by 100 mm.

13.5 Shallow foundations, where accepted after ground improvement, shall be verified for bearing capacity per EN 1997-1 (design approach 2) with the seismic action and liquefaction included, and for uplift by groundwater at grade level; the allowable settlement shall be as given below.

Item	Value	Unit	Status	Basis / remarks
Settlement - buildings, total (long term)	25 mm on piles; 50 mm on improved ground		assumption	REF-CONS-0087
Settlement - angular distortion of buildings and pipe racks	1 / 500		assumption	REF-CONS-0087
Settlement - machine foundations	OEM value; otherwise 1 / 2000 differential along shaft train and 15 mm total		assumption	ER-13.01, REF-CONS-0093
Settlement - storage tanks (LDO and fire water)	per API 650 Annex B (uniform and differential); tank rings on piles		preliminary	REF-CONS-0066

13.6 Settlement of the fill and the compressible clay shall be assessed in a settlement analysis and preloading or waiting periods shall be scheduled before setting out of settlement-sensitive equipment; the pipe connections at the foundation - soil interface (underground piping, cable trenches, duct banks) shall accommodate the relative settlement between piled and grade-supported elements (flexible joints or transition slabs).

14. Drainage, roads and hardstanding

14.1 Surface water drainage shall be designed for the rainfall intensity of DP-CASE-0006 (10 min, 25-year return) using the rational method with runoff coefficients of 0.9 for paved and roofed areas, 0.6 for gravelled areas and 0.3 for landscaped areas; pipe flow depth at design flow not more than 80 % of the diameter and velocities between 0.75 m/s and 4 m/s; the 100-year overland flow path shall be checked to keep floor levels dry.

14.2 Drainage systems shall be segregated: (a) clean rainwater, (b) potentially oily water (transformer bunds, LDO area, workshop, turbine hall floors) through oil separators before discharge, (c) chemical drains to the waste water treatment neutralisation, (d) sanitary. Bunds around LDO tanks, transformers and chemical stores shall contain the largest tank volume plus rainfall of the same intensity for 30 minutes and firewater (per HSE design criteria); the discharge of drainage to the sea shall meet the discharge limits of the EIA (HS-RPT-0001).

14.3 Roads: internal asphalt roads shall be designed for 40 t axle loads to the heaviest delivery route (ER-13.05). The heaviest delivery route shall additionally be checked for the heaviest transport unit of the power island from the quay (400 t quay crane, heavy-lift port 18 km by sea, DP-OWNR-0046); the state road is limited to 120 t gross (DP-OWNR-0045), so transformers, rotors and other loads above this limit shall be shipped by sea and moved on site with self-propelled modular transporters; the route on site shall be checked for the bearing capacity of the fill and of buried services (culverts, duct banks with concrete cover or steel plate).

Item	Value	Unit	Status	Basis / remarks
Pavement structure - heavy roads	60 mm wearing course, 80 mm binder, 300 mm crushed stone base, 300 mm subbase; subgrade CBR $\geq$ 15 %		assumption	ER-13.05; Design life 20 years
Road width and radii	7.0 m for two lanes on the main route; inner turning radius 15 m for heavy transport		assumption	ER-13.05
Vertical clearance over roads (pipe racks, bridges)	not less than 6.0 m; 7.5 m on the heavy transport route		assumption	ER-13.05; Same as the piping design criteria (PI-SPC-0002 section 6)

14.4 Traffic: heavy vehicles shall use the approved routes and time windows of the EIA (no heavy traffic on the village road 07:30 to 09:00 and 16:00 to 18:00); the temporary works and the construction access design shall include this restriction.

15. Marine works basis

15.1 The seawater intake and outfall pipelines shall be of HDPE or GRP with ballast weights and scour protection as required by ER-13.06 and designed for the 50-year wave condition H_s 4.4 m, T_p 8.5 s from the south-west (DP-OWNR-0020). The design flow is 38,000 m³/h (DEC-EPCE-0002); the intake velocity at the screens shall not exceed the EIA value (DP-EPCE-0011) and the water temperature range at the intake is 6 degC to 27 degC (DP-OWNR-0013, DP-OWNR-0012).

15.2 Pipeline stability: the on-bottom stability shall be checked for the 50-year wave and current with the pipeline empty, during installation and operating (safety factor not less than 1.1 against flotation with the empty pipe, not less than 1.3 against sliding and lateral displacement); the pipelines shall be buried or protected by rock cover in the surf zone; scour and the effect of the wave climate on the trench backfill shall be assessed; the design life of the marine pipelines shall be 50 years for HDPE / GRP.

15.3 Marine concrete (intake structure, screen chamber, pumphouse, outfall diffuser blocks) shall be XS3 per section 10; steelwork in the marine zone shall be protected to ISO 12944 category CX (immersion / splash) or be of duplex stainless steel; the screens and rakes shall be designed for the jellyfish load of DP-OWNR-0019 and for the debris load of the storm suspended solids (DP-OWNR-0017).

15.4 Construction constraints: no dredging, trenching or pipe laying between 1 May and 30 September (EIA condition, fish spawning); the marine works design and method statement shall be based on the window 1 October to 30 April, and the tie-in need date of the intake / outfall shall be checked against this window.

16. Buildings and structures

16.1 The buildings are: turbine hall (GT and ST), control and electrical building, GIS building, administration building, workshop / warehouse, water treatment building, gatehouse and fire station (ER-13.02). Steel frames with cladding shall be used for the turbine hall and the workshop; reinforced concrete or composite frames for the control and electrical building; the buildings shall have fire compartments and escape routes per the Fire Regulation and the HSE design criteria (HS-DBR-0001).

16.2 Cladding and roofing: sandwich panels with a coated steel or aluminium skin suitable for C5 (Class RC5 or PVDF-coated, more than 15 years); fixings in stainless steel A4; roof and wall panels of the turbine hall and other buildings with hazardous gas potential shall have blow-out or vent area where required by the explosion protection document (HS-RPT-0005).

16.3 Pipe racks and equipment supports shall be steel frames with the foundation solution of section 13; the design shall provide space for the future post-combustion carbon capture tie-in reservation (ER-14.08) as specified by the layout, without structural rework of the main racks.

16.4 Tanks: the LDO tanks (2 x 3,800 m³, API 650) and fire water tanks (2 x 700 m³) shall be on piled ring walls with the seismic sloshing and base shear per API 650 Annex E adjusted to TBDY 2018 with $I = 1.5$, the freeboard for sloshing being not less than the calculated wave height.

References

Codes and standards cited in this document (Codes, Standards and Regulations Register ALP-EPC-00000-GE-LST-0001; latest edition at contract date unless stated).

Code	Title
Fire Regulation	Regulation on Fire Protection of Buildings (Binalarin Yangından Korunması Hakkında Yonetmelik)
TBDY 2018	Turkish Building Earthquake Code (Türkiye Bina Deprem Yonetmeligi)

Code	Title
API 650	Welded Tanks for Oil Storage
NFPA 850	Recommended Practice for Fire Protection for Electric Generating Plants and HVDC Converter Stations
EN 13001	Cranes - General design (all parts)
EN 1990	Eurocode - Basis of structural design (with EN 1991 actions)
EN 1992	Eurocode 2 - Design of concrete structures
TS 500	Requirements for design and construction of reinforced concrete structures
EN 1993	Eurocode 3 - Design of steel structures
EN ISO 1461	Hot dip galvanized coatings on fabricated iron and steel articles
ISO 12944-2	Paints and varnishes - Corrosion protection of steel structures by protective paint systems - Part 2: Classification of environments
ACI 351.3R	Report on Foundations for Dynamic Equipment
ISO 20816-2	Mechanical vibration - Measurement and evaluation of machine vibration - Part 2: Land-based gas turbines, steam turbines and generators above 40 MW with fluid-film bearings